A Factorial Analysis of Visual-Motor Coordination

The Interacting Effects of Screen Filters and Input Devices

Ducheng Tan

Rochester Institute of Technology
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Motivation and Introduction

- This study examines how screen filter color (Normal, Warm, Gray) and input device type (Mouse, Trackpad) affect visual-motor response time in a point-and-click task. The Human Benchmark Online Reaction Time Tool was used to measure the reaction time of participants in milliseconds.[1]
- The purpose of this experiment is to determine whether screen filter color, input device type, or their combination significantly influences the time a user requires to click on a target stimulus displayed on screen.
- We believe that understanding these effects has practical implications for workspace design and human-computer interaction.

- $n = 30$ total observations across all screen filter and device combinations.
- Trials labeled 1 to 30 and performed in a **randomized order** to minimize learning and fatigue effects.
- One participant at a time while others waited to be called.
- **Controlled conditions:** Same MacBook, portable mouse, reaction time tool, ambient lighting, and location throughout.

Data Collection: Data

Table 1: Response time data (ms) by screen filter and input device

Screen Filter	Device	Obs. 1	Obs. 2	Obs. 3	Obs. 4	Obs. 5
Normal	Mouse	557	590	492	651	456
Normal	Trackpad	688	557	617	689	867
Warm	Mouse	564	504	498	661	465
Warm	Trackpad	661	597	528	729	778
Gray	Mouse	585	561	567	659	430
Gray	Trackpad	567	463	613	709	921

A two way fixed-effects ANOVA model was used to analyze the data. Let Y_{ijk} denote the response time for the k-th replicate under screen filter level i and device level j. The statistical model is given as

$$Y_{ijk} = \mu + \tau_i + \beta_j + (\tau\beta)_{ij} + \varepsilon_{ijk} \tag{1}$$

where μ is the overall mean; τ_i is the main effect of screen filter ($i = 1, 2, 3$ for Gray, Normal, Warm); β_j is the main effect of device ($j = 1, 2$ for Mouse, Trackpad); $(\tau\beta)_{ij}$ is the interaction effect; and ε_{ijk} is the random error term, with $k = 1, \ldots, 5$ replicates per cell. [3]

The model parameter must satisfy the following constrains and assumptions

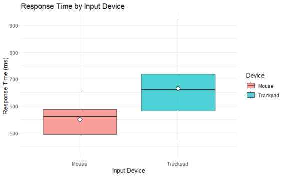
$$\sum_{i=1}^{3} \tau_i = 0, \quad \sum_{j=1}^{2} \beta_j = 0, \quad \sum_{i=1}^{3} (\tau\beta)_{ij} = 0 \ \forall\, j, \quad \sum_{j=1}^{2} (\tau\beta)_{ij} = 0 \ \forall\, i. \tag{2}$$

$$\varepsilon_{ijk} \overset{\text{iid}}{\sim} N(0, \sigma^2). \tag{3}$$

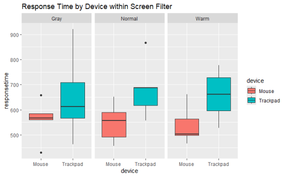
Table 2: Response time (ms) by screen filter and input device

Screen Filter	Device	Mean (ms)	SD (ms)
Normal	Mouse	549.2	77.5
	Trackpad	683.6	116.4
Warm	Mouse	538.4	77.3
	Trackpad	658.6	100.1
Gray	Mouse	560.4	82.7
	Trackpad	654.6	173.2

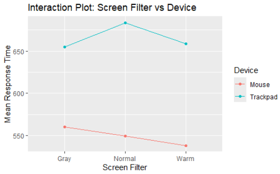
Results: Significant Plots



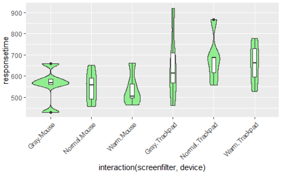
(a) Response time (ms) by input device



(b) Mixed Boxplots



(c) Interaction Plots



(d) Violin Plots

Table 3: Two-way ANOVA table for response time

Source	Df	Sum Sq	Mean Sq	F value	p-value
Screen Filter	2	1,602	801	0.066	0.936
Device	1	101,385	101,385	8.403	0.00788
Screen Filter×Device	2	2,078	1,039	0.086	0.918
Residuals	24	289,571	12,065		

For the main effect of screen filter

$$H_0^{(\tau)} : \tau_1 = \tau_2 = \tau_3 = 0 \qquad H_1^{(\tau)} : \text{at least one } \tau_i \neq 0. \tag{4}$$

For the main effect of device

$$H_0^{(\beta)} : \beta_1 = \beta_2 = 0 \qquad H_1^{(\beta)} : \text{at least one } \beta_j \neq 0. \tag{5}$$

For the interaction we test

$$H_0^{(\tau\beta)} : (\tau\beta)_{ij} = 0 \quad \forall i, j \qquad H_1^{(\tau\beta)} : \text{at least one } (\tau\beta)_{ij} \neq 0. \tag{6}$$

The main effect of input device is significant ($F_{1,24} = 8.40$, $p = 0.00788$), indicating different response times for Mouse and Trackpad. The effects of screen filter ($F_{2,24} = 0.066$, $p = 0.936$) and the interaction ($F_{2,24} = 0.086$, $p = 0.918$) are not significant. Thus, we reject $H_0^{(\beta)}$ for device but fail to reject $H_0^{(\tau)}, H_0^{(\tau\beta)}$ for screen filter and the interaction.

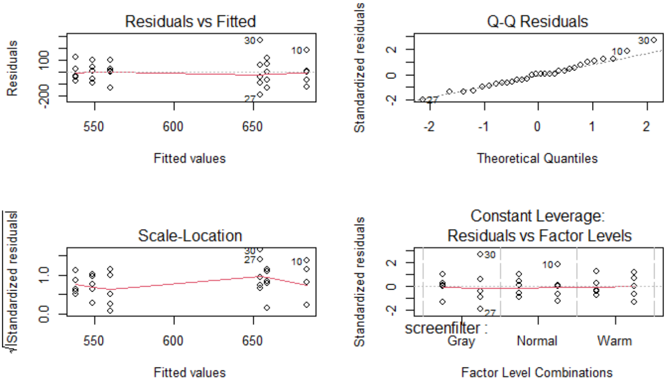


Figure 1: Diagnostic plots for the fitted two-way ANOVA model [2]

Results: Assumptions *continued*

- The Q-Q Residuals plot: standardized residuals follow the theoretical normal quantiles, Shapiro Wilk' Test with a test statistic of $W = 0.97541$ and a p-value of 0.6947, therefore we have no evidence to reject that the residuals are not normally distributed.
- Equal Variance: The Residuals vs. Fitted plot shows no systematic patter, Levene's test was used to obtain $F(5, 24) = 0.6175$ and a p-value of 0.6876 to demonstrate statistically that the group variance are equal.
- The Scale-Location plot: spread of residuals is approximately constant across fitted values
- The Constant Leverage: Residuals vs. Factor Levels plot shows no unusual leverage or outlying observations.

Results: Tukey HSD

Calculating the Tukey Threshold for each effects yields that only the device main effect is significant

$$|\bar{Y}_{.\text{Trackpad}.} - \bar{Y}_{.\text{Mouse}.}| = 116.2667 > 82.7808 = T_{0.05} \qquad (7)$$

```
$device
                    diff      lwr      upr     p adj
Trackpad-Mouse 116.2667 33.4859 199.0474 0.0078829
```

Figure 2: Significant Tukey Difference for $T_{0.05} = 82.7808$

Conclusions and Limitations

- Input device is the only significant factor ($F_{1,24} = 8.40$, $p = 0.008$)
- Trackpad responses $\approx$ 116 ms slower
- Screen filter type not significant ($F_{2,24} = 0.066$, $p = 0.936$)
- No significant interaction effect ($F_{2,24} = 0.086$, $p = 0.918$)
- Model assumptions satisfied:
  - Normality ($W = 0.975$, $p = 0.695$)
  - Homoscedasticity ($F(5, 24) = 0.618$, $p = 0.688$)
  - Independence
- Limitation: Trials conducted sequentially (potential bias from waiting effects)
- Future improvement: include participants as a blocking factor

References

[1] Human Benchmark. Aim trainer—user profile. https://humanbenchmark.com/users/69dd6e920f1d1c143b377887/aim. Online reaction time tool. Accessed April 2026.

[2] Joseph McKean, Robert V Hogg, and Allen Craig. *Introduction to mathematical statistics*. 2019.

[3] Douglas C Montgomery. *Design and analysis of experiments*. John wiley & sons, 2017.

Thank you for listening!